

Dash Robots

WHAT'S IN THE KIT

- 5 Dash robots
- 1 set of Dash/Dot coding cards
- Charging cables



QUICK-START GUIDE

1. Charge Dash before class (micro-USB).
2. Connect via the free Blockly or Wonder app on a tablet/Chromebook.
3. Start with Drive mode to learn controls, then switch to coding.
4. Use the coding cards for unplugged planning before app time.

FULL LESSONS IN THIS GUIDE

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PRO TIP: Five robots = perfect station rotation for a full class.

Dash Robots

LESSON · GRADES K-2

Dash Delivers

LEARNING OUTCOMES

- Students will program a robot to complete a delivery route
- Students will break a journey into segments
- Students will work in assigned team roles

MATERIALS

- Dash robots (5 stations)
- Tablets/Chromebooks with Blockly or Wonder app
- Delivery 'cargo' (notes, small flat objects) and station signs
- Tape roads between stations

INSTRUCTIONS

1. Free-drive for 5 minutes first — get the 'make it move' excitement out.
2. Set up delivery stations around the room with tape roads.
3. Teams program one delivery leg at a time: drive, turn, arrive, sound the horn.
4. Roles rotate each leg: programmer, cargo loader, route checker.
5. Chain the legs into a full route with deliveries at each stop.
6. Celebrate with a full-route run while the class tracks each delivery.

ACTIVITY

Dash Delivers — a delivery robot with real cargo and stations. Programming one leg at a time teaches decomposition the gentle way, and the horn at each stop never gets old.

Dash Robots

LESSON · GRADES 3-5

Robot Obstacle Rally

LEARNING OUTCOMES

- Students will use sensor blocks to make a robot react to obstacles
- Students will debug programs systematically
- Students will score accuracy over speed in evaluation

MATERIALS

- Dash robots + devices with Blockly
- Obstacle course: taped path, cardboard walls, tunnel
- Accuracy scoring sheet (touches, off-path moments)
- Debugging log

INSTRUCTIONS

1. Course walkthrough: teams map the course on paper first.
2. Build v1 programs with movement blocks; run and score touches.
3. Introduce the obstacle sensor block: 'when wall detected, do...'
4. Teams add reactive behavior and log every change in the debugging log.
5. Scored rally runs: fewest touches wins, not fastest time.
6. Debrief the logs: what was your best find-and-fix?

ACTIVITY

Robot Obstacle Rally — accuracy-scored runs make careful debugging beat reckless speed. The sensor block turns 'follow my commands' into 'react to your world' — a big conceptual leap.